

180. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

180.1 Overview

This clause specifies the 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 Physical Medium Dependent (PMD) types together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. These PMD types provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links over one, two, four, or eight pairs, respectively, of single-mode fiber, with a reach of up to at least 500 m.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 180–1 through Table 180–4, to the medium through the medium dependent interface (MDI).

Table 180–1—Physical Layer clauses associated with the 200GBASE-DR1 PMD

Associated clause	200GBASE-DR1
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120C—200GAUI-8 C2M ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120E—200GAUI-4 C2M ^c	Optional ^d
120F—200GAUI-2 C2C ^c	Optional ^d
120G—200GAUI-2 C2M ^c	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
176D—200GAUI-1 C2M	Optional ^d

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.5.1.

^c When implemented in a 200GBASE-DR1 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 200GAUI-n may be instantiated within a 200GBASE-DR1 PHY as described in 176B.5.1.

Table 180–2—Physical Layer clauses associated with the 400GBASE-DR2 PMD

Associated clause	400GBASE-DR2
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120C—400GAUI-16 C2M ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120E—400GAUI-8 C2M ^c	Optional ^d
120F—400GAUI-4 C2C ^c	Optional ^d
120G—400GAUI-4 C2M ^c	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
176D—400GAUI-2 C2M	Optional ^d

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.6.1.

^c When implemented in a 400GBASE-DR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 400GAUI-n may be instantiated within a 400GBASE-DR2 PHY as described in 176B.6.1.

Table 180–3—Physical Layer clauses associated with the 800GBASE-DR4 PMD

Associated clause	800GBASE-DR4
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a

^a One or two 800GAUI-n may be instantiated within a 800GBASE-DR4 PHY as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

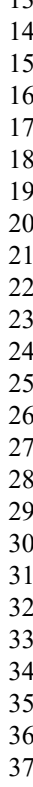
^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Table 180–4—Physical Layer clauses associated with the 1.6TBASE-DR8 PMD

Associated clause	1.6TBASE-DR8
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
120G—1.6TAUI-16 C2M	Optional ^a
170—1.6Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII Extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a
176D—1.6TAUI-8 C2M	Optional ^a

^a One or two 1.6TAUI-n may be instantiated within a 1.6TBASE-DR8 PHY as described in 176B.8.1.

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180.3 PMD service interface

This subclause specifies the services provided by the PMD. The service interface for these PMDs are described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (PMA immediately above the PMD). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and 174.3 for 1.6TBASE-R. For PMD:IS_UNITDATA_*i*.request primitive in the transmit direction and PMD:IS_UNITDATA_*i*.indication primitive in the receive direction, where *i* = 0 to *n*−1, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 106.25 GBd. The number of parallel streams, *n*, is 1 for 200GBASE-DR1, 2 for 400GBASE-DR2, 4 for 800GBASE-DR4, and 8 for 1.6TBASE-DR8.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive (see 116.3.3.3) corresponds to the variable rts_status of the ready-to-send (RTS) function (see 180.5.12), as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_*i*.indication primitive on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the PMD:IS_SIGNAL.request primitive (see 116.3.3.4).

180.4 Delay and Skew

180.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-DR1 PMD including 2 m of fiber in one direction shall be no more than 14 848 bit times (29 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR2 PMD including 2 m of fiber in one direction shall be no more than 29 696 bit times (58 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-DR4 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 1.6TBASE-DR8 PMD including 2 m of fiber in one direction shall be no more than 118 784 bit times (232 pause_quanta or 74.24 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 for 200GBASE-DR1 and 400GBASE-DR2, in 169.4 for 800GBASE-DR4, and in 174.4 for 1.6TBASE-DR8.

180.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS.

180.4.2.1 Skew constraints for 200GBASE-DR1 and 400GBASE-DR2

Skew and Skew Variation for 200GBASE-DR1 and 400GBASE-DR2 are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–4 and Figure 116–5. For 200GBASE-DR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 200GBASE-DR1 and 400GBASE-DR2 see 116.5.

180.4.2.2 Skew constraints for 800GBASE-DR4 and 1.6TBASE-DR8

Skew and Skew Variation for 800GBASE-DR4 are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

Skew and Skew Variation for 1.6TBASE-DR8 are defined in 174.5 and specified at the points SP1 to SP6 shown in Figure 174–5 and Figure 174–6.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 800GBASE-DR4 see 169.5 and for 1.6TBASE-DR8 see 174.5.

180.5 PMD functional specifications

The 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

180.5.1 Block diagram

The block diagram for 800GBASE-DR4 is shown in Figure 180–2.

The block diagram for 200GBASE-DR1 is equivalent to Figure 180–2, but for one lane per direction, instead of four.

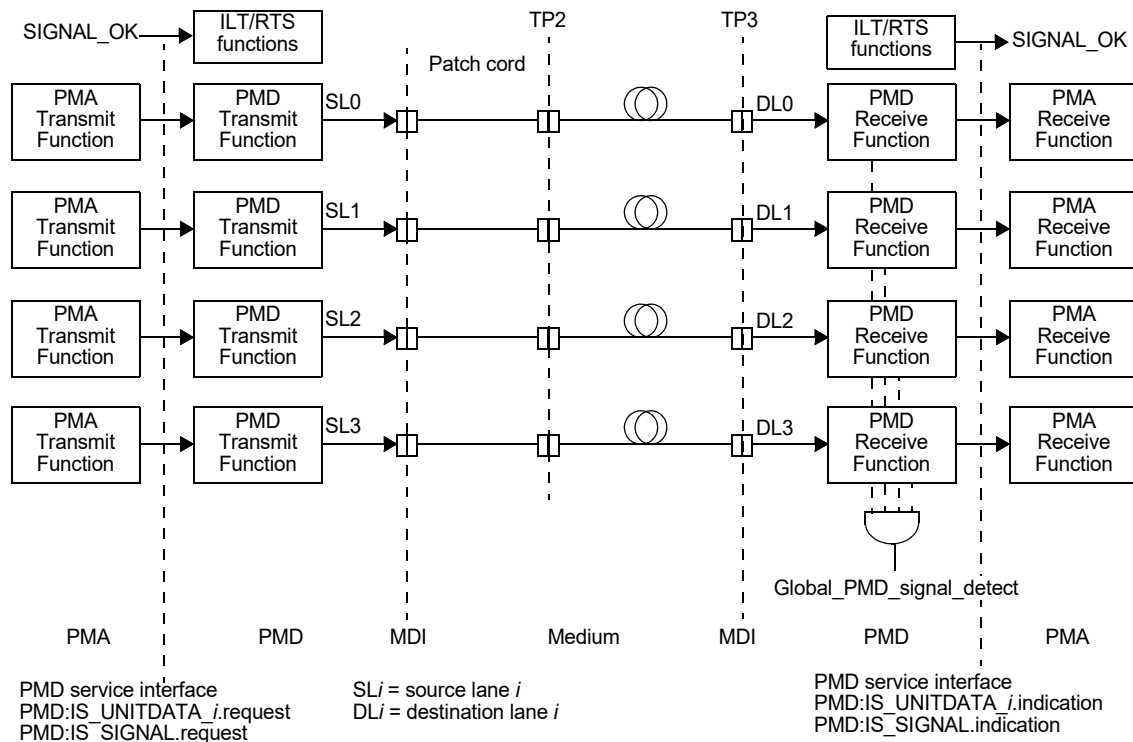
The block diagram for 400GBASE-DR2 is equivalent to Figure 180–2, but for two lanes per direction, instead of four.

The block diagram for 1.6TBASE-DR8 is equivalent to Figure 180–2, but for eight lanes per direction, instead of four.

The inter-sublayer link training (ILT) and RTS functions indicated in Figure 180–2 are defined in Annex 178B.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 180.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 180.8.3). Unless specified otherwise, all receiver measurements and tests defined in 180.9 are made at TP3.

While TP2 and the patch cord used for transmitter measurements and tests are shown in Figure 180–2, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.



NOTE: For clarity, only one direction of transmission is shown

Figure 180–2—Block diagram for 800GBASE-DR4 transmit/receive paths

180.5.2 PMD transmit function

The PMD transmit function has two operating modes: DATA and TRAINING. The operating mode is controlled by the tx_mode variable of the ILT function (see 180.5.12): it is DATA when tx_mode = DATA, and TRAINING otherwise.

When operating in DATA mode, the PMD transmit function shall convert the n symbol streams requested by the PMD service interface PMD:IS_UNITDATA_0.request primitive to PMD:IS_UNITDATA_ $n-1$.request primitive into n separate optical signals. The n optical signals shall then be delivered to the MDI, which contains n parallel light paths for transmit, according to the transmit optical specifications in this clause.

The polarity of the PMD output on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI output, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI output, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 178B.7.7). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

When operating in TRAINING mode, the PAM4 symbol stream on each lane is provided by the ILT function (see 178B.7).

180.5.3 PMD receive function

The PMD Receive function shall convert the n parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the PMD:IS_UNITDATA_0.indication primitive to PMD:IS_UNITDATA_ $n-1$.indication primitive, all according to the receive optical specifications in this clause.

The polarity of the PMD input on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI input, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI input, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.7.7). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

180.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all n lanes. For the 200GBASE-DR1 PMD, the variable Global_PMD_signal_detect is set to PMD_signal_detect_0 (see 180.5.5). For the 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMDs, the variable Global_PMD_signal_detect is set to the logical-and of PMD_signal_detect_ i from all of the PMD lanes.

180.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable PMD_signal_detect_ i to either 0 or 1 according to the criteria in Table 180–5.

Table 180–5—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_i
Average optical power at TP3 ≤ -15 dBm	0
[The input signal is a compliant 200GBASE-R, 400GBASE-R, 800GBASE-R or 1.6TBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 180–8]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_i variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

180.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the ILT and RTS functions associated with the PMD (see 180.5.12).

180.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the Global_PMD_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 180–7.

If a PMD_fault is detected, then the PMD may set the Global_PMD_transmit_disable variable to one, turning off the optical transmitter in each lane.

180.5.8 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function allows the optical transmitters in each lane to be selectively disabled. This function is used by the ILT function. It may optionally be provided through control variables PMD_transmit_disable_i.

If either PMD_transmit_disable_i or the variable tx_disable in lane i of the ILT function (see 178B.8.2.1) is set to one (where i represents the lane number in the range 0:n – 1), this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 180–7.

If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_i variable to one, turning off the optical transmitter in each lane.

180.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault variable to one.

180.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

180.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

180.5.12 Autonomous path startup (APSU) functions

A PMD shall provide the ready-to-send (RTS) function and the ILT function with a training frame of O1 format, specified in Annex 178B.

When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

When tx_mode = LOCAL_PATTERN (see 178B.8.3.1) the PMD transmits PRBS31Q (see 176.7.4.2).

The default value of max_wait_timer_duration is 60, corresponding to a duration of 60 seconds for max_wait_timer (see 178B.8.3.3).

180.6 Lane assignments

In order to support the ILT function as described in 180.5.12, optical lane assignments (within a group of transmit or receive lanes) for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 are specified. The positioning and ordering of transmit and receive lanes at the MDI is specified in 180A.4.

Each fiber between the transmitter of one PHY and the receiver of another PHY shall connect to the same signal lane number. For example, a fiber connects SL1 at the transmitting end to DL1 at the receiving end.

180.7 PMD to MDI optical specifications

The operating range for the 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMD is defined in Table 180–6. A 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers according to the specifications defined in Table 180–11. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-DR2 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Table 180–6—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 operating range

PMD type	Required operating range
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8	2 m to 500 m

180.7.1 Transmitter optical specifications

A transmitter shall meet the specifications defined in Table 180–7 per the definitions in 180.9.

Table 180–7—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 transmit characteristics

Description	Reference	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Signaling rate, each lane (range)	—	106.25 ± 50 ppm		GBd
Modulation format	—	PAM4		—
Lane wavelength (range)	180.9.3	1304.5 to 1317.5		nm
Side-mode suppression ratio (SMSR), each lane (min)	180.9.3	30		dB
Average launch power, each lane (max)	180.9.4	4		dBm
Average launch power, each lane ^a (min)	180.9.4	−3.1 ^b		dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (max)	180.9.5	4.2		dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (min) for max(TECQ, TDECQ) < 0.9 dB for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB	180.9.5	−0.1 −1 + max(TECQ, TDECQ)		dBm dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	180.9.6	3.4		dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	180.9.7	3.4		dB
TDECQ – TECQ , each lane (max)	—	2.5		dB
Transmitter functional symbol error histogram, each lane (max)	180.9.8	See Table 180–17		—
Transmitter overshoot and undershoot, each lane (max)	180.9.9	27		%
Transmitter power excursion, each lane (max)	180.9.10	2.9		dBm

Table 180–7—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 transmit characteristics (*continued*)

Description	Reference	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Extinction ratio, each lane (min)	180.9.11	3.5		dB
Transmitter transition time, each lane (max)	180.9.12	8		ps
Average launch power of OFF transmitter, each lane (max)	—	–15		dBm
RIN_x OMA, each lane (max) x = 15.5 x = 21.4	180.9.13	–139 —	— –139	dB/ Hz dB/ Hz
Optical return loss tolerance, each lane (max)	—	15.5	21.4	dB
Transmitter reflectance ^c , each lane (max)	—	–26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of –3.1 dBm corresponds to an OMA of –0.1 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 180–8 vary with $\max(TECQ, TDECQ)$. The relationships are illustrated in Figure 180–3 along with the values for OMA_{outer} each lane (max).

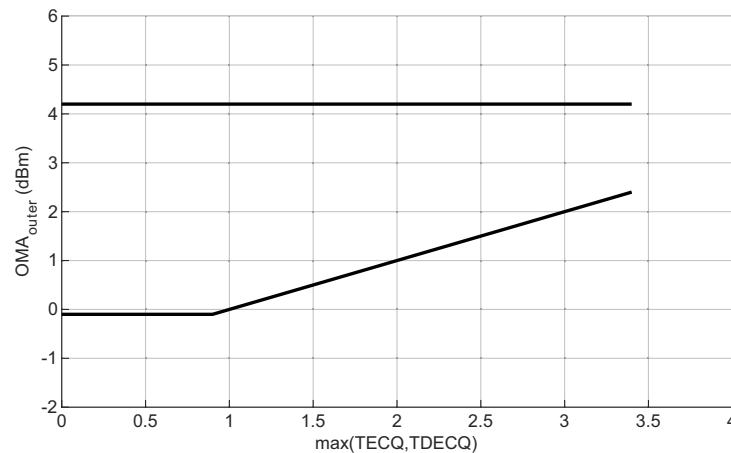


Figure 180–3— OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus $\max(TECQ, TDECQ)$

180.7.2 Receive optical specifications

A receiver shall meet the specifications defined in Table 180–8 per the definitions in 180.9.

Table 180–8—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 receive characteristics

Description	Reference	Value	Unit
Signaling rate, each lane (range)	—	106.25 ± 50 ppm	GBd
Modulation format	—	PAM4	—
Lane wavelengths (range)	180.9.3	1304.5 to 1317.5	nm
Damage threshold ^a , each lane	—	5	dBm
Average receive power, each lane (max)	—	4	dBm
Average receive power, each lane ^b (min)	—	−6.1	dBm
Receive power (OMA _{outer}), each lane (max)	—	4.2	dBm
Receiver reflectance (max)	—	−26	dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max) for TECQ < 0.9 dB for 0.9 dB ≤ TECQ ≤ SECQ	180.9.14	−3.4 −4.3 + TECQ	dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	180.9.15	−0.9	dBm
Conditions of stressed receiver sensitivity test ^d :			
Stressed eye closure for PAM4 (SECQ), lane under test	—	3.4	dB
OMA _{outer} of each aggressor lane ^e	—	4.2	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 180.8) for the BLER specified in 180.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^e No aggressors needed for 200GBASE-DR1 in a single lane device.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 180–8 vary depending on the value of TECQ for the test source used for measurement. Figure 180–4 illustrates this.

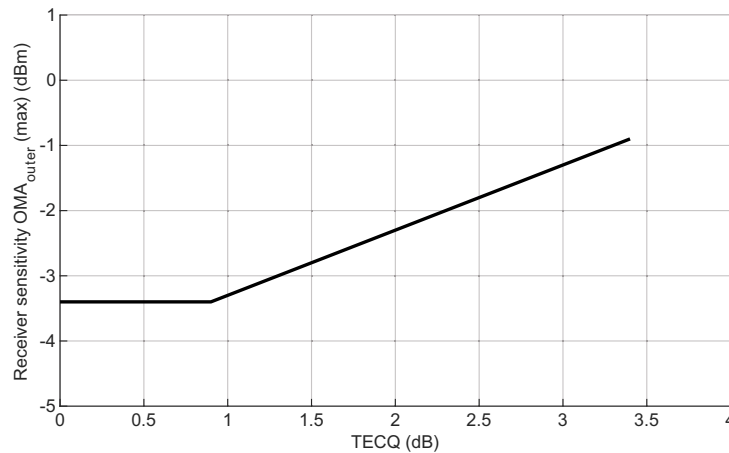


Figure 180-4—Receiver sensitivity (OMA_{outer}), each lane (max)

180.7.3 Illustrative link power budget

An illustrative power budget and penalties for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 channels are shown in Table 180-9.

Table 180-9—Illustrative link power budget

Parameter	Value	Unit
Power budget (for max TDECQ)	6.7	dB
Operating distance	500	m
Channel insertion loss ^{a, b}	3	dB
Maximum discrete reflectance	-35	dB
Allocation for penalties ^c (for max TDECQ)	3.7	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 180-6 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 180.8.2.

^b The channel insertion loss of 3 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.1 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.5 dB depending on the discrete reflectances in the link as shown in Table 180-12.

^c Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. In this illustrative link power budget the value shown includes an allocation of 0.1 dB for MPI penalty and 0.2 dB for DGD penalty. For cases with a channel insertion loss less than 3 dB, as shown in Table 180-12, the allocation for penalties should be “6.7 – channel insertion loss”.

The values of transmitter OMA_{outer} (max), transmitter OMA_{outer} (min) versus max (TECQ, TDECQ), and receiver sensitivity (OMA_{outer}) (max) versus TECQ are illustrated in Figure 180-5.

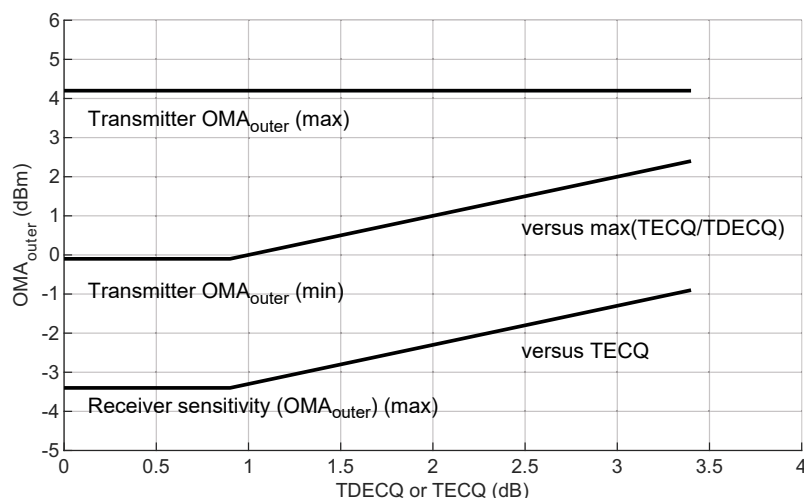


Figure 180-5—Transmitter OMA_{outer} each lane versus $\max(TECQ, TDECQ)$ and receiver sensitivity (OMA_{outer}) each lane versus TECQ

180.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 180-6.

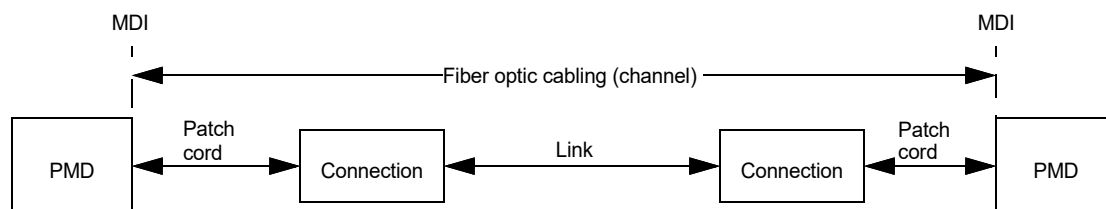


Figure 180-6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 180-10. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. The maximum value of channel insertion loss is dependent on the number and maximum value of the discrete reflectances within the channel as given in Table 180-12. Discrete reflectances below -55 dB may be ignored when determining the supported channel insertion loss.

Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 series one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 180–10—Optical channel characteristics

Description	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Operating distance (max)	500		m
Channel insertion loss ^{a,b} (max)	See Table 180–12		dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	0.65		ps/nm
Negative dispersion ^b (min)	–0.85		ps/nm
DGD_max ^c	2.24		ps
Optical return loss (min)	27	37	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

180.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables meeting the characteristics in Table 180–11. The use of optical fiber cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), type G.657.A1, or type G.657.A2 (bend insensitive) fibers is recommended.

Table 180–11—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in TIA 568-C.3.

180.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 180–6, consists of a mated pair of optical connectors.

The maximum link distance is based on an allocation of 0.25 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 180–12 being allocated to connection

and splice loss. For example, with no discrete reflections greater than -55 dB, a total connection and splice loss allocation of 2.75 dB supports five connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 180–10 are met.

Table 180–12—Maximum channel insertion loss versus number of discrete reflectances

Maximum channel insertion loss (dB)		Number of discrete reflectances > −55 dB and ≤ −45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectances > −45 dB and ≤ −35 dB	0	3	3	3	3	3	3	2.9	2.9	2.9
	1	3	3	3	2.9	2.9	2.9	2.9	2.8	2.8
	2	2.9	2.9	2.9	2.8	2.8	2.8	2.7	2.7	2.7
	3	2.8	2.8	2.7	2.7	2.7	2.6	2.6	2.6	2.6
	4	2.7	2.6	2.6	2.6	2.5	2.5	2.5	2.5	— ^a
	5	2.5	2.5	2.5	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a
	6	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

180.8.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 180–6). The PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 180A–1. Example constructions of the MDI include the following:

- PMD with a connectorized fiber pigtail plugged into an adapter
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 180.5.1, not at the MDI.

Annex 180A specifies the details of the MDIs for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8.

For 200GBASE-DR1, besides the option to connect to an MDI with two fibers, there are two additional specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 400GBASE-DR2, and 800GBASE-DR4 there are two specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 1.6TBASE-DR8 there is one specified MDI optical receptacle, a single-row 16 fiber interface.

180.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

180.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 180–14 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 180–14 may be used to perform that test. The test patterns used in this clause are shown in Table 180–13.

Table 180–13—Test patterns

Pattern	Pattern description	Defined in
3 ^a	PRBS31Q	176.7.4.2
4	PRBS13Q	176.7.4.3
5 ^a	Scrambled idle	119.2.4.9, 172.2.4.11, 175.2.4.11
6	SSPRQ	176.7.4.5
7	Valid 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R signal	119.2.4, 172.2.4, 175.2.4

^a Precoding (see 176.7.1.2) is added to this pattern as necessary (see 180.9.14 and 180.9.15).

Table 180–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	3, 4, 5, 6, or 7	180.9.3
Side mode suppression ratio	3, 5, 6, or 7	180.9.3
Average optical power	3, 5, 6, or 7	180.9.4
Outer optical modulation amplitude (OMA _{outer})	4 or 6	180.9.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	180.9.6
Transmitter eye closure for PAM4 (TECQ)	6	180.9.7
Transmitter functional test	3	180.9.8
Transmitter overshoot and undershoot	6	180.9.9
Transmitter power excursion	6	180.9.10
Extinction ratio	4 or 6	180.9.11

Table 180–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Transmitter transition time	6	180.9.12
RIN_x OMA	4 or 6	180.9.13
Receiver sensitivity	3	180.9.14
Stressed receiver conformance test signal calibration	6	180.9.15
Stressed receiver sensitivity	3	180.9.15

180.9.2 Reference receiver

A reference receiver is defined for transmitter measurements. The reference receiver is composed of an O/E converter, a clock recovery unit (CRU), and a pattern-triggered oscilloscope as shown in Figure 180–7. The combined frequency response of the O/E converter and oscilloscope has a 3 dB bandwidth of approximately 53.125 GHz with a fourth-order Bessel-Thomson response for frequencies up to at least 1.3×106.25 GHz and a response no higher than –20 dB for frequencies greater than 1.3×106.25 GHz.

The CRU has a corner frequency of 4 MHz and a slope of 20 dB/decade. The CRU may be implemented in hardware or software depending on oscilloscope technology.

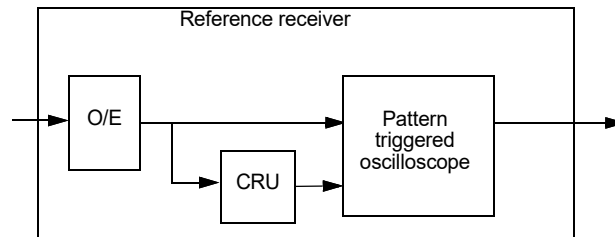


Figure 180–7—Reference receiver block diagram

180.9.3 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 180–7 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns defined in Table 180–14.

180.9.4 Average optical power

The average optical power of each lane shall be within the limits given in Table 180–7 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns defined in Table 180–14.

180.9.5 Outer optical modulation amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limit given in Table 180–7. The OMA_{outer} is measured using one of the test patterns specified for OMA_{outer} in Table 180–14 as the difference between the average optical

launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 180–8. OMA_{outer} is measured using the waveforms captured at the output of the reference receiver defined in 180.9.2.

When measured with pattern 6, OMA_{outer} is measured as the difference between the average optical launch power level P_3 , measured over the central 2 UI of the first 7 UI of all runs of 7 or more threes, and the average optical launch power level P_0 , measured over the central 2 UI of the first 6 UI of all runs of 6 or more zeros.

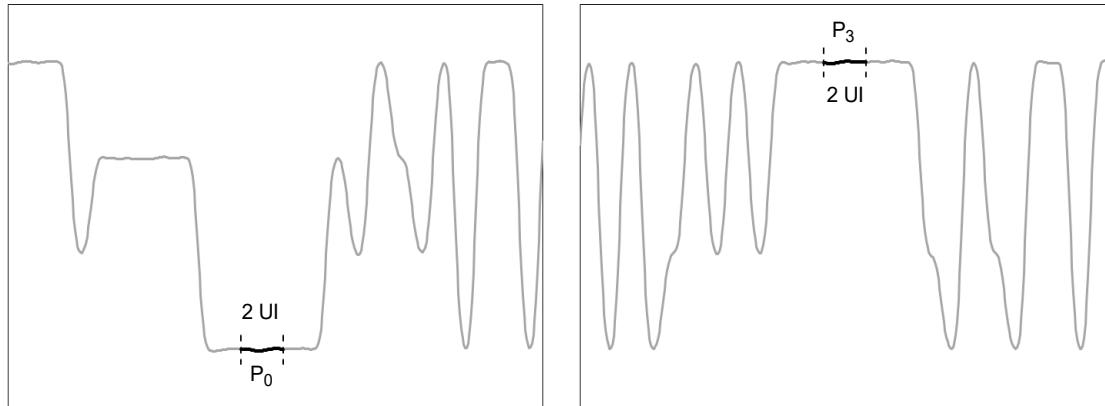


Figure 180–8—Example power levels P_0 and P_3 from PRBS13Q test pattern

180.9.6 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limits given in Table 180–7.

TDECQ is a measure of each optical transmitter vertical eye closure when transmitted through a worst case optical channel specified in 180.9.6.2, as measured through a reference receiver specified in 180.9.2, and equalized with the reference equalizer as described in 180.9.6.3. The reference equalizer may be implemented in software or may be part of an oscilloscope.

180.9.6.1 TDECQ conformance test setup

A block diagram for the TDECQ conformance test is shown in Figure 180–9. Other equivalent measurement implementations may be used with suitable calibration.

Each optical lane is tested individually with all other transmit lanes in operation and all lanes using the same test pattern. There shall be at least 31 UI delay between the test pattern on one lane and the pattern on any other lane, so that the symbols on each lane are not correlated within the PMD. TDECQ is measured with all receive lanes of the PMD in operation using pattern 3 or 5 (see Table 180–13). The received test patterns shall be asynchronous to the pattern used to test the transmitter and shall have power levels as specified in Table 180–8 for the aggressor lanes in the stressed receiver sensitivity test.

The test pattern is transmitted repetitively by the optical lane under test and the oscilloscope is set up to capture the complete pattern for TDECQ analysis as described in 180.9.6.4.

For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 180–1 through Table 180–4), the clock source for the transmitted SSPRQ test pattern is derived from

the clock recovered from the xAUI-n input signal. The pattern of the xAUI-n input signal may be PRBS31Q, scrambled idle, or a valid xBASE-R signal. The lane under test transmits the SSPRQ test pattern instead of the test pattern from the xAUI-n input signal.

The optical splitter and variable reflector are adjusted so that each transmitter is tested with the optical return loss specified in Table 180–7. The state of polarization of the back reflection is adjusted to create the greatest RIN. Each optical lane is tested with the optical channel described in 180.9.6.2.

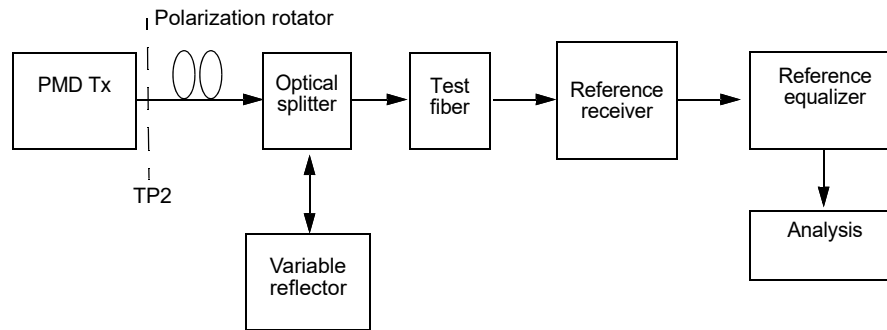


Figure 180–9—TDECQ conformance test block diagram

180.9.6.2 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 180–15.

Table 180–15—Transmitter compliance channel specifications

Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
Minimum	Maximum			
$0.0463 \times (\lambda - 1311) - 0.55$	$0.0443 \times (\lambda - 1311) + 0.37$	Minimum	See Table 180–7	0.5

^a Dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 500 m distance. The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024).

^b The value “Minimum” implies that the test channel insertion loss does not excessively degrade the signal to noise ratio of the measurement.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 180–15 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 180–15. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 180–15.

180.9.6.3 Reference equalizer

The reference equalizer is a T-spaced, discrete-time equalizer with 15 feed-forward taps and a single decision feedback tap, where T is the symbol period. Equalizer coefficient constraints are given in Table 180–16. The reference equalizer may be implemented in the oscilloscope.

The equalizer is modeled as shown in Figure 180–10, where

- $r(t)$ is the output of the reference receiver defined in 180.9.2
- $\eta(t)$ is colored Gaussian noise, generated from additive white Gaussian noise (AWGN) filtered by a Bessel-Thomson (BT) filter as defined in 180.9.2
- $z(t)$ is the input signal given by Equation (180–1)
- z_n is the sampled response of input signal $z(t)$, given by Equation (180–2)
- y_n is the equalizer output corresponding to x_n
- x_n is the n th symbol in the test pattern sequence
- w is the set of feed-forward tap coefficients, given by Equation (180–3)
- $b(1)$ is the feedback tap coefficient
- the feedback is modeled as correct, i.e., x_n is the pattern used to generate the input $r(t)$
- the received signal $r(t)$ is aligned with the test pattern such that $r(nT)$ corresponds to x_n

$$z(t) = r(t) + \eta(t) \quad (180-1)$$

$$z_n = z(nT + \phi), \text{ where } 0 \leq \phi < T \quad (180-2)$$

$$w = \{w(a), w(a+1), \dots, w(a+14)\} \quad (180-3)$$

where the variable a is an integer in the range -3 to 0 associated with the permitted number of feed-forward pre-cursor taps as specified in Table 180–16.

NOTE—This reference equalizer is part of the test and does not imply any particular receiver equalizer architecture in a compliant implementation.

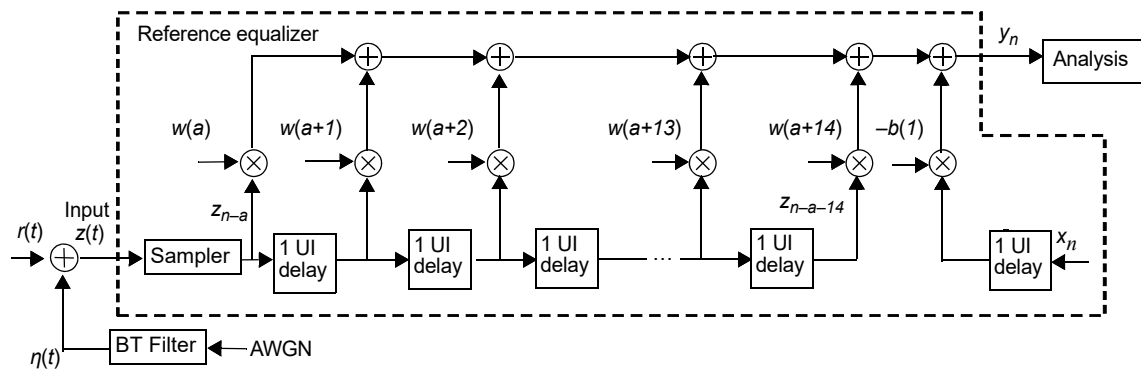


Figure 180–10—TDECQ reference equalizer functional model

Table 180–16—Reference equalizer tap coefficients

Parameter	Symbol	Value	
		Minimum	Maximum
Number of feed-forward taps	N_w	15	
Number of pre-cursor feed-forward taps	—	0	3
Main tap coefficient limit	$w(0)$	$0.9 - b(1)$	2.5
Normalized feed-forward tap coefficient limits: $i = -3$ $i = -2$ $i = -1$ $i = 1$ $i = 2$ $i = 3$ $i = 4$ $i = 5$ $i = 6$ $i \geq 7$	$w(i)/w(0)$	-0.15 -0.1 -0.5 -0.6 -0.2 -0.15 -0.15 -0.15 -0.15 -0.1	0.1 0.25 0.1 0.2 0.3 0.15 0.15 0.15 0.15 0.1
Pre-post-cursor tap coefficient difference limit: $ w(1)/w(0) - b(1) - w(-1)/w(0) $	—	—	0.25
Feed-forward tap DC gain ^a	—	1	
Number of feedback taps	N_b	1	
Feedback tap coefficient limit ^b	$b(1)$	0	0.33

^a The sum of all 15 feed-forward tap coefficients, $w(i)$.

^b The feedback tap coefficient $b(1)$ is referenced to $\text{OMA}_{\text{TDECQ}}/2$ (see 180.9.6.4 for definition of $\text{OMA}_{\text{TDECQ}}$).

180.9.6.4 TDECQ measurement method

The standard deviation of the noise of the reference receiver specified in 180.9.2, σ_S , is determined with no optical input signal and the same settings as used to capture samples described below.

The test pattern specified for TDECQ (see Table 180–14) is transmitted repetitively by the optical lane under test and the oscilloscope is set up to capture samples from all symbols in the complete pattern without averaging.

If an equivalent-time sampling oscilloscope is used, the impact of the sampling process and the reference equalizer on transmitter noise has to be compensated for, so that the correct magnitude of noise is present at the output of the equalizer.

The captured waveform is processed to find the largest noise that could be combined with the signal by a reference receiver when optimally equalized by a reference equalizer. The optimal equalizer tap coefficients are dependent on the amount of noise that can be added to the signal, so finding the noise that can be added and the optimal equalizer setting is an iterative process. One way of doing this, using estimated PAM4 symbol error ratio as the figure of merit for the equalized signal, is described below.

The reference equalizer specified in 180.9.6.3 is applied to the captured waveform, $r(t)$.

For a given sampling phase ϕ_0 and assumed RMS value σ_G for the colored Gaussian noise $\eta(t)$, determine the equalizer coefficients $\mathbf{w}$ and $b(1)$, subject to the limits in Table 180–16, that result in the minimum mean squared error (MMSE) between y_n and x_n . Alternative optimization methods may be used to determine equalizer tap weights if they report comparable values of TDECQ.

An eye diagram to help illustrate TDECQ is shown in Figure 180–11. The eye diagram corresponds to the optimized equalizer setting at sampling phase ϕ_0 . The feedback symbol x_n changes at phase $\phi = \phi_0 - T/2$.

NOTE—The colored Gaussian noise $\eta(t)$ added to $r(t)$ in Figure 180–10 is not reflected in the eye diagram. It is accounted for separately when estimating the SER in the analysis block.

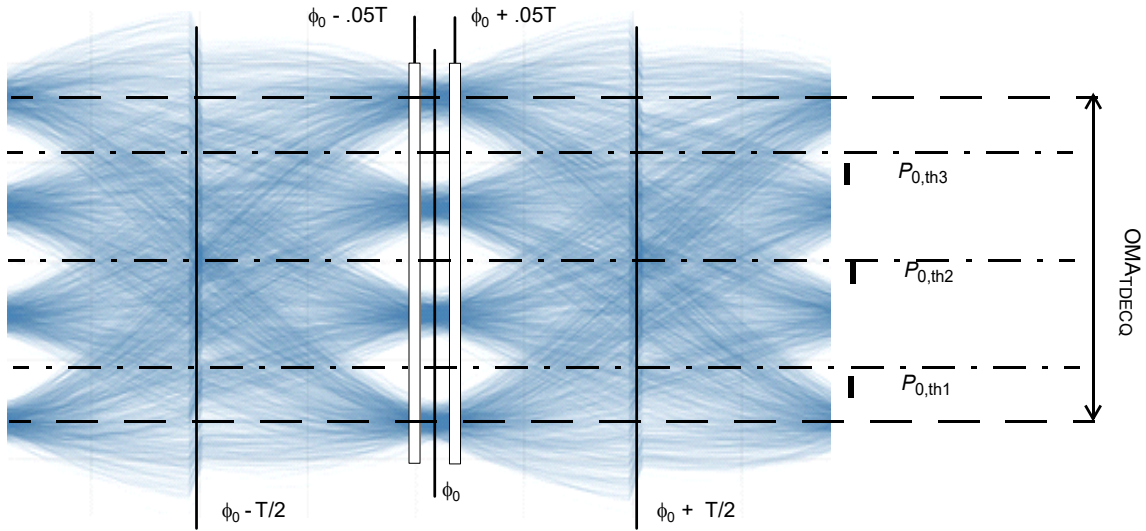


Figure 180–11—Illustration of the TDECQ measurement

For a given sampling phase ϕ_0 two regions of a UI are measured, centered 0.05 UI before and after the sampling phase ϕ_0 , each with a width of 0.04 UI. For the left region, all samples such that $\phi_0 - 0.07T \leq \phi \leq \phi_0 - 0.03T$ are included. For the right region, all samples such that $\phi_0 + 0.03T \leq \phi \leq \phi_0 + 0.07T$ are included. The precise time position of the pair of regions is adjusted to minimize TDECQ while keeping the regions spaced 0.1 UI apart. Let $y_{n,L}$ and $y_{n,R}$ be the equalized samples in the left region and the right region respectively.

The nominal threshold levels $P_{0,th1}$, $P_{0,th2}$, and $P_{0,th3}$, are determined from the OMA_{TDECQ} and the average optical power P_{ave} as defined in Equation (180–4), Equation (180–5), and Equation (180–6), and illustrated in Figure 180–11.

$$P_{0,th1} = P_{ave} - \frac{OMA_{TDECQ}}{3} \quad (180-4)$$

$$P_{0,th2} = P_{ave} \quad (180-5)$$

$$P_{0,th3} = P_{ave} + \frac{OMA_{TDECQ}}{3} \quad (180-6)$$

where OMA_{TDECQ} is measured on the equalized captured waveform y_n using the method defined in 180.9.5.

The samples captured in each region are processed to, in effect, combine the PAM4 waveform with noise, in order to produce an estimate of the PAM4 symbol error ratio (SER). One way of doing this is described below.

σ_G quantifies the amount of noise that can be added at the receiver while still achieving the target SER. For a given value of σ_G , the symbol error ratio for the left region, SER_L , and that for the right region, SER_R , are estimated as described below using the tap coefficients optimized to minimize the mean squared error between y_n and x_n at ϕ_0 .

The probability that a sample value in the left region, $y_{n,L}$, will become a symbol error given the addition of noise $\eta(t)$ with RMS value of σ_G is defined in Equation (180–7). The probability that a sample value in the right region, $y_{n,R}$, will become a symbol error given the addition of noise $\eta(t)$ with RMS value of σ_G is defined in Equation (180–7) except that $y_{n,L}$ is replaced with $y_{n,R}$.

$$P(y_{n,L}, \sigma_G) = \frac{1}{2} \begin{cases} \operatorname{erfc}\left(\frac{P_{th1} - y_{n,L}}{\sqrt{2} C_{eq} \sigma_G}\right) & x_n = 0 \\ \operatorname{erfc}\left(\frac{y_{n,L} - P_{th1}}{\sqrt{2} C_{eq} \sigma_G}\right) + \operatorname{erfc}\left(\frac{P_{th2} - y_{n,L}}{\sqrt{2} C_{eq} \sigma_G}\right) & x_n = 1 \\ \operatorname{erfc}\left(\frac{y_{n,L} - P_{th2}}{\sqrt{2} C_{eq} \sigma_G}\right) + \operatorname{erfc}\left(\frac{P_{th3} - y_{n,L}}{\sqrt{2} C_{eq} \sigma_G}\right) & x_n = 2 \\ \operatorname{erfc}\left(\frac{y_{n,L} - P_{th3}}{\sqrt{2} C_{eq} \sigma_G}\right) & x_n = 3 \end{cases} \quad (180-7)$$

where

C_{eq} is a coefficient which accounts for the reference equalizer noise enhancement

The value of C_{eq} can be calculated from the product of the normalized noise power density spectrum $N(f)$ at the input of the reference equalizer and the normalized frequency response $H_{eq}(f)$ of the feed-forward section of the reference equalizer, as shown in Equation (180–8).

$$C_{eq} = \sqrt{\int_f N(f) \times |H_{eq}(f)|^2 df} \quad (180-8)$$

where

$N(f)$ is the normalized noise power density spectrum equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 53.125 GHz.

and

$$\int_f N(f) df = H_{eq}(f=0) = 1 \quad (180-9)$$

The SER for the left region, SER_L , is defined by Equation (180–10). The SER for the right region, SER_R , is defined by Equation (180–11).

$$SER_L(\sigma_G) = \frac{1}{N} \sum_n P(y_{n,L}, \sigma_G) \quad (180-10)$$

where N is the number of all samples in the left region.

$$SER_R(\sigma_G) = \frac{1}{N} \sum_n P(y_{n,R}, \sigma_G) \quad (180-11)$$

where N is the number of all samples in the right region.

When the SER is estimated, the actual thresholds, P_{th1} , P_{th2} , and P_{th3} , are adjusted by up to $\pm 2\%$ of OMA_{TDECQ} from $P_{0,th1}$, $P_{0,th2}$, and $P_{0,th3}$ to minimize $\max(SER_L(\sigma_G), SER_R(\sigma_G))$. If $\max(SER_L, SER_R)$ is not equal to the target PAM4 SER of 4.56×10^{-4} , the process is iterated by changing σ_G and reoptimizing the equalizer tap coefficients. TDECQ is calculated using the optimized σ_G that results in $\max(SER_L, SER_R)$ as close as practical to the target PAM4 SER.

The RMS noise, R , that could be added by a receiver is given by Equation (180-12).

$$R = \sqrt{\sigma_G^2 + \sigma_S^2} \quad (180-12)$$

At a given sampling phase ϕ_0 , TDECQ(ϕ_0) is given by Equation (180-13).

$$TDECQ(\phi_0) = 10 \log_{10} \left(\frac{OMA_{outer}}{6} \times \frac{1}{Q_t R} \right) \quad (180-13)$$

where

OMA_{outer} is measured as defined in 180.9.5
 Q_t is 3.428, consistent with the target symbol error ratio for Gray mapped PAM4, and can be calculated according to Equation (180-14)

$$Q_t = \sqrt{2} \operatorname{erfc}^{-1} \left(\frac{4}{3} SER_{target} \right) \quad (180-14)$$

TDECQ is the minimum value of TDECQ(ϕ_0) calculated over $0 \leq \phi_0 < T$.

180.9.7 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter eye closure at TP2. The TECQ shall be within the limit given in Table 180-7 if measured using the test pattern specified for TECQ in Table 180-14. TECQ is measured using the methods specified for TDECQ in 180.9.6, except that the test fiber is not used.

180.9.8 Transmitter functional test (TFT)

The transmitter functional test uses a functional receiver (FnRx) defined in 180.9.8.1 to measure a transmitter functional symbol error histogram defined in 180.9.8.2. A block diagram showing the FnRx is shown in Figure 180-12. The transmitter under test is connected to the FnRx by a test fiber, which meets the requirements in 180.9.6.2. Full compliance with 180.9.6.2 requires the test fiber to provide back reflection with polarization state resulting in the greatest RIN. Other test fibers, including patch cords, may be used for additional tests.

For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 180–1 through Table 180–4), the clock for the transmitter output is recovered from the xAUI-n input signal.

The transmitter under test is configured with precoding set appropriately for the FnRx being used in the test. The appropriate precoding state may be communicated via the ILT function or by other means.

The transmitter block error histograms are measured using the method described in 174A.9.3.

A probable failure is indicated by either of the following:

- exceeding the transmitter functional symbol error histogram mask defined in Table 180–17
- a value greater than 0 in counter `test_block_error_bin_i_k` (see 174A.9.2), for k greater than 8

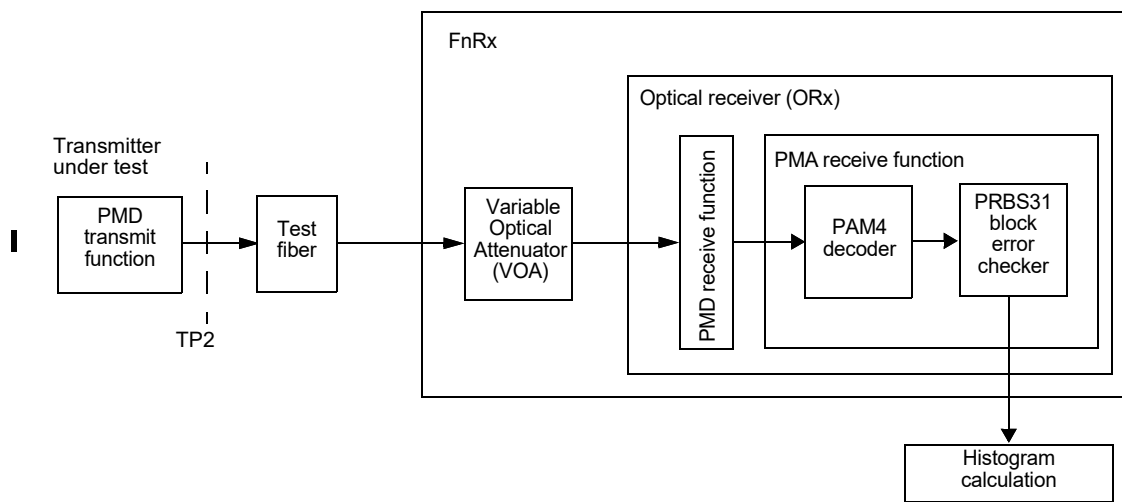


Figure 180–12—Transmitter functional test block diagram

180.9.8.1 FnRx definition

The FnRx is a variable optical attenuator (VOA) followed by an optical receiver (ORx) that complies with the characteristics in Table 180–8. The VOA and test fiber approximate the compliance channel as given in Table 180–15 with the exception that the channel loss is not minimum. The VOA level is set to the value given by Equation (180–15).

NOTE 1—The intent of the first and second terms of Equation (180–15) is to calibrate differences in test fibers and optical receivers, respectively, to reduce variability across different test conditions.

$$VOA_level = Test_fiber_correction + ORx_TECQ_correction - Test_margin \quad (180-15)$$

where:

- Test_fiber_correction* is given by Equation (180–16), and is the difference between the power budget used to determine the transmitter under test OMA_{outer} (min) and the best estimate of the test fiber power budget
- ORx_TECQ_correction* is the difference between receiver sensitivity (max) as given in Table 180–8 and ORx receiver sensitivity, both at transmitter under test TECQ, and is given by Equation (180–19)

Test_margin is the additional *ORx_OMA* to improve SNR of the transmitter functional test. It is equal to 1.5 dB. The ORx operating BER target assuming random errors is changed to 2.4×10^{-5} . The transmitter functional symbol error mask (see Table 180–17) is calculated using this BER target.

$$Test_fiber_correction = Tx_DUT_power_budget - Test_fiber_power_budget \quad (180-16)$$

where:

Tx_DUT_power_budget is the transmitter under test power budget as specified in Table 180–9, except that the TDECQ value measured over the test fiber is used instead of the max TDECQ value. It is given by Equation (180–17)

Test_fiber_power_budget is the power budget of the test fiber using the best estimates of the test fiber channel insertion loss, MPI and DGD penalties, and TDECQ of the DUT over the test fiber and is given by Equation (180–18)

$$Tx_DUT_power_budget = Channel_insertion_loss + MPI_and_DGD_penalties_allocation + \max(DUT_TECQ, DUT_TDECQ) \quad (180-17)$$

where:

Channel_insertion_loss is the “Channel insertion loss” as given in Table 180–9

MPI_and_DGD_penalties_allocation

is the allocation for MPI and DGD penalties as given in Table 180–9

DUT_TECQ is the measured TECQ for the transmitter under test

DUT_TDECQ is the measured TDECQ for the transmitter under test

$$Test_fiber_power_budget = Test_fiber_loss + Test_fiber_MPI_and_DGD_penalties + Test_fiber_DUT_TDECQ \quad (180-18)$$

where:

Test_fiber_loss is the best estimate of the actual channel insertion loss of the test fiber

Test_fiber_MPI_and_DGD_penalties

is the best estimate of the actual MPI and DGD penalties of the test fiber

Test_fiber_DUT_TDECQ is the best estimate of the transmitter under test actual TDECQ value over the test fiber. An example estimate is the measured *DUT_TECQ* value plus the best estimate of the actual transmitter under test CD penalty over the test fiber

$$ORx_TECQ_correction = RxS_OMA_max_at_DUT_TECQ - ORx_RxS_OMA_at_DUT_TECQ \quad (180-19)$$

where:

RxS_OMA_max_at_DUT_TECQ

is the receiver sensitivity OMA_{outer} (max) spec at the measured TECQ of the transmitter under test

ORx_RxS_OMA_at_DUT_TECQ

is the ORx receiver sensitivity OMA_{outer} at the measured TECQ of the transmitter under test

NOTE 2—The correction factor given by *ORx_TECQ_correction* might be moderately pessimistic depending upon the relative characteristics of the device used as the reference receiver and the transmitter under test. A failure of the TFT accurately identifies a transmitter that is not generally interoperable. However, the test might falsely fail a generally interoperable transmitter, which might be addressed by a moderate increase in transmitter OMA_{outer} to pass TFT.

180.9.8.2 Transmitter functional symbol error histogram

The transmitter functional symbol error histogram mask for each lane is given in Table 180–17. The limit $H_{\max}(k)$ is calculated using the method in 174A.9.5 using $BER_{\max}$ (ORx operating BER target) of 2.4×10^{-5} and $p = 1$. ORx operating BER target corresponds to *Test_margin* increase from the receiver sensitivity of ORx. This improves the measurement SNR of the transmitter functional test. ORx and *Test_margin* are defined in 180.9.8.1.

The transmitter functional symbol error histogram is measured using a FnRx as defined in 180.9.8.1, the method defined in 174A.9.3, and the test pattern given in Table 180–14. Minimum measurement time is 60 s.

Table 180–17—Transmitter functional symbol error mask

Test symbol errors per test block, k (see 174A.9.5)	Probability $H_{\max}(k)$
1	1.15×10^{-1}
2	7.47×10^{-3}
3	3.24×10^{-4}
4	1.05×10^{-5}
5	2.73×10^{-7}
6	5.88×10^{-9}
7	1.08×10^{-10}
8	1.75×10^{-12}

180.9.9 Transmitter overshoot and undershoot

The transmitter overshoot and undershoot of each lane shall be within the limit given in Table 180–7 if measured using the test pattern specified for overshoot and undershoot in Table 180–14.

Transmitter overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 180.9.6) and the waveform captured for the TECQ test (see 180.9.7), at the output of the reference receiver defined in 180.9.2.

Transmitter overshoot and undershoot are calculated using the methods in 140.7.7, with the exceptions that P_0 , P_3 and OMA_{outer} are measured using the methods in 180.9.5 and the hit ratio used to determine $P_{\max}$ and $P_{\min}$ is 1×10^{-4} .

180.9.10 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 180–7 if measured using the test pattern specified for transmitter power excursion in Table 180–14.

Transmitter power excursion is measured using the waveforms captured at the output of the reference receiver defined in 180.9.2.

Transmitter power excursion is defined in 140.7.8, with the exception that the hit ratio used to determine $P_{\max}$ and $P_{\min}$ is 1×10^{-4} .

180.9.11 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 180–7 if measured using one of the test patterns specified for extinction ratio in Table 180–14. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power levels P_3 and P_0 , as defined for OMA_{outer} (see 180.9.5).

180.9.12 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 180–7 if measured using the test pattern specified for transmitter transition time in Table 180–14.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer}, or from 80% of OMA_{outer} to 20% of OMA_{outer}, for the rising and falling edges respectively. The transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 180.9.2.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 180.9.5). When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

180.9.13 Relative intensity noise (RIN_xOMA)

RIN_xOMA of each lane, with “x” referring to the value for optical return loss tolerance in Table 180–7, shall be within the limit given in Table 180–7 when measured using one of the test patterns and sampling range specified for OMA_{outer} measurement in 180.9.5, but with applied x dB optical return loss and the reference receiver defined in 180.9.2. The noise is measured at the output of the reference receiver.

RIN_xOMA is calculated as defined in Equation (180–20).

$$\text{RIN}_x\text{OMA} = 10\log_{10}\left[\frac{(N_0 + N_3)^2}{\text{OMA}_{\text{outer}}^2 B}\right] \quad (180-20)$$

where:

RIN _x OMA	is the relative intensity noise referred to optical modulation amplitude, measured with x dB optical return loss
N_3	is the RMS deviation from the mean of the optical power at the P_3 level
N_0	is the RMS deviation from the mean of the optical power at the P_0 level
OMA _{outer}	is the optical modulation amplitude in milliwatts
B	is the noise bandwidth of the reference receiver (Hz)

N_0 and N_3 are each measured in milliwatts in a histogram window placed so that the slope of the waveform across the histogram window is negligible. These might be 1 UI wide or narrower, and they are expected to give lower readings for N_0 or N_3 than other positions or wider histogram windows.

NOTE—The noise bandwidth of a 4th-order Bessel-Thomson filter is approximately 1.04 times its 3 dB bandwidth.

180.9.14 Receiver sensitivity

The receiver sensitivity ($\text{OMA}_{\text{outer}}$) of each lane shall be within the limits given in Table 180–8 if measured using one of the test patterns specified for receiver sensitivity in Table 180–14. The conformance test signal at TP3 meets the requirements for a transmitter followed by an attenuator.

The TECQ of the conformance test signal is measured according to 180.9.7. The measured value of TECQ is then used to calculate the limit for receiver sensitivity ($\text{OMA}_{\text{outer}}$) as specified in Table 180–8.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

The measured receiver sensitivity is the lowest value of $\text{OMA}_{\text{outer}}$ where the PMD receiver meets the BLER requirements in 180.2, measured at the PMA using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 180–19. The error mask $H_{\text{max}}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_{\text{m}}(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_{\text{m}}(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_{\text{m}}(k)) - \log_{10}(H_{\text{m}}(4))) / (k - 4)$ should be less than $(\log_{10}(H_{\text{m}}(4)) - \log_{10}(H_{\text{m}}(2))) / 2$ for each k greater than 4.

The Low-stressed Receiver Sensitivity (LRS) conformance test signal at TP3 meets the transmit characteristics given in Table 180–7 except for the characteristics in Table 180–18.

Table 180–18—LRS conformance test signal exceptions to transmit characteristics

Parameter	Value	Unit
TECQ (max)	1.8	dB
TECQ (min)	1.4	dB
Overshoot and undershoot (max)	10	%
Extinction ratio (max)	5.5	dB
Extinction ratio (min)	4.5	dB
Transition time (max)	6.5	ps
Transition time (min)	4.5	ps
ORL tolerance (max)	15.5	dB
Average launch power of each non test signal lane (max)	-15	dBm

Table 180–19—Error ratio parameters

Parameter name	Value
p 200GBASE-DR1 400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	1 2 4 8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	6.4×10^{-5}
BER_{max}	2.281×10^{-4}

Table 180–20—Receiver error mask

Test symbol errors per test block, k (see 174A.9.5)	Probability $H_{\text{max}}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
1	3.6×10^{-1}	3.3×10^{-1}	2.3×10^{-1}	1.3×10^{-1}
2	2.2×10^{-1}	1×10^{-1}	3.5×10^{-2}	1×10^{-2}
3	9.2×10^{-2}	2.1×10^{-2}	3.6×10^{-3}	5.1×10^{-4}
4	2.8×10^{-2}	3.3×10^{-3}	2.7×10^{-4}	1.9×10^{-5}
5	7×10^{-3}	4×10^{-4}	1.6×10^{-5}	5.5×10^{-7}
6	1.4×10^{-3}	4.1×10^{-5}	8.2×10^{-7}	1.3×10^{-8}
7	2.5×10^{-4}	3.5×10^{-6}	3.5×10^{-8}	2.7×10^{-10}
8	3.9×10^{-5}	2.7×10^{-7}	1.3×10^{-9}	4.7×10^{-12}
9	5.3×10^{-6}	1.8×10^{-8}	4.2×10^{-11}	7.1×10^{-14}
10	6.4×10^{-7}	1.1×10^{-9}	1.2×10^{-12}	9.6×10^{-16}
11	7.1×10^{-8}	5.8×10^{-11}	3.2×10^{-14}	1.2×10^{-17}
12	7.2×10^{-9}	2.9×10^{-12}	7.5×10^{-16}	1.3×10^{-19}
13	6.8×10^{-10}	1.3×10^{-13}	1.6×10^{-17}	1.2×10^{-21}
14	5.9×10^{-11}	5.6×10^{-15}	3.3×10^{-19}	1.1×10^{-23}
15	4.7×10^{-12}	2.2×10^{-16}	6.1×10^{-21}	9.1×10^{-26}
16	3.6×10^{-13}	8.1×10^{-18}	1×10^{-22}	6.9×10^{-28}

180.9.15 Stressed receiver sensitivity

Stressed receiver sensitivity of each lane shall be within the limit given in Table 180–8 if measured using the method defined in 121.8.10 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 180.9.6, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 180–7.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the RIN_{xOMA} of the SRS test source should be no greater than the value specified in Table 180–7.
- The frequency of the sinusoidal interferer is greater than 200 MHz.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 180–7 using one of the test patterns specified in Table 180–14.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ OMA_{outer} of each aggressor lane” are as given in Table 180–8.

The measured stressed receiver sensitivity is the lowest value of OMA_{outer} where the receiver meets the requirements in 180.2 using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 180–19. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

180.10 Safety, installation, environment, and labeling

180.10.1 General safety

All equipment subject to this clause shall conform to IEC 62368-1.

180.10.2 Laser safety

200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.⁹

⁹A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

180.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

180.10.4 Environment

Normative specifications in this clause are met by a system integrating a 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

180.10.5 Electromagnetic emission

A system integrating a 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

180.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

180.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-DR4).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 180.10.2.

180.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 180–21 and Table 180–22. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 180–21 and Table 180–22.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 180–21—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	180.5.6	1.0.15	45.2.1.1
Global_PMD_transmit_disable	180.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_7 to PMD_transmit_disable_4 ^a	180.5.8	1.9.8 to 1.9.5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	180.5.8	1.9.4, 1.9.3	45.2.1.8
PMD_transmit_disable_1 ^c	180.5.8	1.9.2	45.2.1.8
PMD_transmit_disable_0	180.5.8	1.9.1	45.2.1.8

^a Available only in 1.6TBASE-DR8

^b Available only in 800GBASE-DR4 and 1.6TBASE-DR8

^c Available only in 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

Table 180–22—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	180.5.9	1.1.7	45.2.1.2
PMD_transmit_fault	180.5.10	1.8.11	45.2.1.7
PMD_receive_fault	180.5.11	1.8.10	45.2.1.7
Global_PMD_signal_detect	180.5.4	1.10.0	45.2.1.9
PMD_signal_detect_7 to PMD_signal_detect_4 ^a	180.5.5	1.10.8 to 1.10.5	45.2.1.9
PMD_signal_detect_3, PMD_signal_detect_2 ^b	180.5.5	1.10.4, 1.10.3	45.2.1.9
PMD_signal_detect_1 ^c	180.5.5	1.10.2	45.2.1.9
PMD_signal_detect_0	180.5.5	1.10.1	45.2.1.9

^a Available only in 1.6TBASE-DR8

^b Available only in 800GBASE-DR4 and 1.6TBASE-DR8

^c Available only in 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

180.12 Protocol implementation conformance statement (PICS) proforma for Clause 180, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8¹⁰

180.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 180, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

180.12.2 Identification

180.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

180.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-20xx, Clause 180, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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¹⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

180.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*DR1	200GBASE-DR1	180.7	Device supports requirements for 200GBASE-DR1 PHY	O.1	Yes [] No []
*DR2	400GBASE-DR2	180.7	Device supports requirements for 400GBASE-DR2 PHY	O.1	Yes [] No []
*DR4	800GBASE-DR4	180.7	Device supports requirements for 800GBASE-DR4 PHY	O.1	Yes [] No []
*DR8	1.6TBASE-DR8	180.7	Device supports requirements for 1.6TBASE-DR8 PHY	O.1	Yes [] No []
*INS	Installation / cable	180.8	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
DC	Delay constraints	180.4.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	180.4.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	180.11	Registers and interface supported	O	Yes [] No []

180.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

180.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PCS and PMA	180.1		M	Yes []
F2	Block error ratio	180.2	Meets the BLER specified in 180.2	M	Yes []
F3	Transmit function	180.5.2	Conveys symbols from PMD service interface to MDI	M	Yes []
F4	Mapping between optical signal and logical signal for transmitter	180.5.2	Optical power levels from lowest to highest correspond to tx_symbols zero, one, two, and three, respectively	M	Yes []
F5	Receive function	180.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F6	Conversion of four optical signals to symbol streams	180.5.3	For delivery to the PMD service interface	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F7	Mapping between optical signal and logical signal for receiver	180.5.3	Optical power levels from lowest to highest correspond to rx_symbols zero, one, two, and three, respectively	M	Yes []
F8	Global Signal Detect behavior	180.5.4	Global_PMD_signal_detect is an indicator of the presence of an optical signal on all lanes	M	Yes []
F9	Lane-by-lane Signal Detect function	180.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 180–5	M	Yes []
F10	PMD reset function	180.5.6	Resets the PMD sublayer	O	Yes [] No []
F11	Global transmit disable function	180.5.7	Disables all of the optical transmitters with the Global_PMD_transmit_disable variable	O	Yes [] No []
F12	PMD lane-by-lane transmit disable using management variable	180.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	O	Yes [] No []
F13	PMD lane-by-lane transmit disable using state variable	180.5.8	Disables the optical transmitter when tx_disable variable is true	M	Yes []
F14	PMD_fault function	180.5.9	Sets PMD_fault to one if any local fault is detected	O	Yes [] No []
F15	PMD_transmit_fault function	180.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	O	Yes [] No []
F16	PMD_receive_fault function	180.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	O	Yes [] No []

180.12.4.2 PMD to MDI optical specifications for 200GBASE-DR1

Item	Feature	Subclause	Value/Comment	Status	Support
DDR1	Transmitter meets specifications in Table 180–7	180.7.1	Per definitions in 180.9	DR1:M	Yes [] N/A []
DDR2	Receiver meets specifications in Table 180–8	180.7.2	Per definitions in 180.9	DR1:M	Yes [] N/A []

180.12.4.3 PMD to MDI optical specifications for 400GBASE-DR2

Item	Feature	Subclause	Value/Comment	Status	Support
EDR1	Transmitter meets specifications in Table 180–7	180.7.1	Per definitions in 180.9	DR2:M	Yes [] N/A []
EDR2	Receiver meets specifications in Table 180–8	180.7.2	Per definitions in 180.9	DR2:M	Yes [] N/A []

180.12.4.4 PMD to MDI optical specifications for 800GBASE-DR4

Item	Feature	Subclause	Value/Comment	Status	Support
FDR1	Transmitter meets specifications in Table 180–7	180.7.1	Per definitions in 180.9	DR4:M	Yes [] N/A []
FDR2	Receiver meets specifications in Table 180–8	180.7.2	Per definitions in 180.9	DR4:M	Yes [] N/A []

180.12.4.5 PMD to MDI optical specifications for 1.6TBASE-DR8

Item	Feature	Subclause	Value/Comment	Status	Support
GDR1	Transmitter meets specifications in Table 180–7	180.7.1	Per definitions in 180.9	DR8:M	Yes [] N/A []
GDR2	Receiver meets specifications in Table 180–8	180.7.2	Per definitions in 180.9	DR8:M	Yes [] N/A []

180.12.4.6 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	180.9	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	180.9.3	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	180.9.4	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	180.9.5	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	180.9.6	Each lane	M	Yes []
OM6	Transmitter eye closure for PAM4 (TECQ)	180.9.7	Each lane	M	Yes []
OM8	Transmitter functional test (TFT)	180.9.8	Each lane	M	Yes []
OM8	Transmitter overshoot and undershoot	180.9.9	Each lane	M	Yes []
OM9	Transmitter power excursion	180.9.10	Each lane	M	Yes []
OM10	Extinction ratio	180.9.11	Each lane	M	Yes []
OM11	Transmitter transition time	180.9.12	Each lane	M	Yes []
OM12	Relative intensity noise	180.9.13	Each lane	M	Yes []
OM13	Receiver sensitivity	180.9.14	Each lane	M	Yes []
OM14	Stressed receiver sensitivity	180.9.15	Each lane	M	Yes []

180.12.4.7 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	180.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	180.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	180.10.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

180.12.4.8 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	180.8	Meets requirements specified in Table 180–10 and Table 180–11	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	180.8.2	Meets requirements specified in Table 180–12	INS:M	Yes [] N/A []
OC3	MDI Requirements	180.8.3	Meets IEC 61753-1 and IEC 61753-021-2	INS:M	Yes [] N/A []

180.12.4.9 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	180.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	180.11		MD:M	Yes [] N/A []